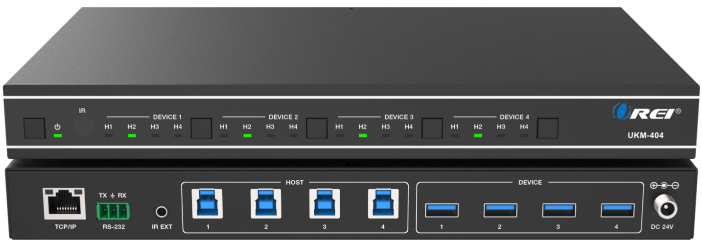

UKM-404

USB 3.2 Gen 1 4x4 Matrix

USER MANUAL

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Table of Contents

Thank you for purchasing this product

For optimum performance and safety, please read these instructions carefully before connecting, operating or adjusting this product. Please keep this manual for future reference.

Surge protection device recommended

This product contains sensitive electrical components that may be damaged by electrical spikes, surges, electric shock, lightning strikes, etc. Use of surge protection systems is highly recommended in order to protect and extend the service life of your equipment.

Registration Page

Please activate your warranty by registering your product through the link below - **www.orei.com/register**

Technical Support

Need Help?
Our experienced Technical Support Team is here for you to answer your questions, give technical advice or help troubleshoot your project to get you installed on time and on budget. Call, email or chat with us now.

OREI Live Technical Support Hours
US team (US/Canada/Mexico): Monday-Friday, 9 a.m. - 5 p.m. Central Time
Support Email - info@orei.com | **Support Number** - 877-290-5530

Or

Chat Live on www.orei.com
Send us an instant message now. Our Technical Support Team will respond momentarily. Available during live support hours.

Introduction

The UKM-404 USB Matrix enables seamless sharing of 4 USB peripherals among 4 computers or laptops. Compliant with USB 3.2 Gen 1, it is also backward compatible with USB 2.0 and 1.1. It supports data transfer speeds of up to 5Gbps and delivers 5V/1.5A power on all USB A ports. The device is plug-and-play, requiring no drivers, and works with Windows, Linux, and MacOS. Control is simplified via the included remote, RS-232, or WebGUI, making it perfect for personal and commercial use.

Features & Package Contents

Features

1. USB 3.2 Gen 1 compliant

2. Data transfer rate up to 5Gbps

3. Connect and switch between 4 USB devices on 4 computers/laptops

4. 5V/1.5A output on USB-A ports

5. Backwards compliant with USB 2.0 and 1.1

6. Plug and Play – no drivers needed

7. Compatible with Windows, Linux and MacOS

8. Simplified control with included remote, RS-232 and WebGUI

Package Contents

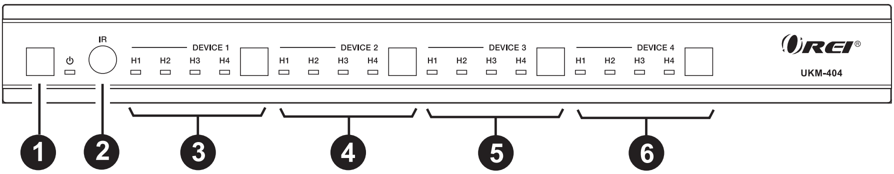
1.	UKM-404	1pcs
2.	IR Remote	1pcs
3.	IR Receiver Cable (1.5m)	1pcs
4.	USB Cable (USB 3.0 Type A to Type B, 1.8m)	4pcs
5.	3pin-3.81mm Phoenix Connector	1pcs
6.	Machine Screw (KM3*4)	4pcs
7.	Mounting Ear	2pcs
8.	Power Adapter	1pcs
9.	User Manual	1pcs

Specifications

Technical	
USB Protocol	USB 3.2 Gen 1
Data Transfer Rate	5Gbps
IR Level	5Vp-p
IR Frequency	Fixed 38KHz
Transmission Distance	3m/9.8ft for USB 3.2 Gen 1 5Gbps over passive cable
ESD Protection	IEC 61000-4-2: ±8kV (Air-gap discharge) & ±4kV (Contact discharge)
Connection	
Inputs	4 × USB Host [USB 3.2 Gen 1 Type B, 9-pin female]
Outputs	4 × USB Device [USB 3.2 Gen 1 Type A, 9-pin female]
Control	1 × TCP/IP [RJ45] 1 × RS-232 [3pin-3.81mm phoenix connector] 1 × IR EXT [3.5mm stereo mini-jack]
Mechanical	
Housing	Metal Enclosure
Color	Black
Dimensions	L: 270mm / 10.62in | W: 166mm / 6.53in | H: 30mm / 1.18in
Weight	1.16kg / 2.55lbs
Power Supply	Input: AC 100-240V 50/60Hz, Output: DC 24V/2A (US/EU standard, CE/FCC/UL certified)
Power Consumption	13.2W (Max)
Operating Temperature	32 - 104˚F / 0 - 40˚C
Storage Temperature	-4 - 140˚F / -20 - 60˚C
Operating Humidity	20%~80% relative humidity, non-condensing
Storage Humidity	10%~90% relative humidity, non-condensing

Operation Controls and Functions

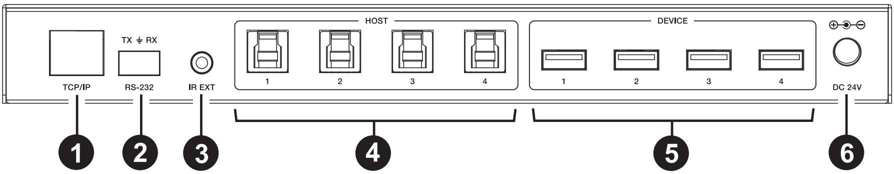
Front Panel



No.	Name	Function Description
1.	Power button & indicator	Press the button once to power on the matrix, the Green LED will light up; Long press the button for 3 seconds to set it to standby mode, the Red LED will light up.
2.	IR	IR signal receiver, receiving the signal from the IR remote.
3.	DEVICE 1 button & H1~H4 indicator	Press the button to switch between Hosts(H1~H4) circularly for device 1. When the host is selected, the corresponding led will light up.
4.	DEVICE 2 button & H1~H4 indicator	Press the button to switch between Hosts(H1~H4) circularly for device 2. When the host is selected, the corresponding led will light up.
5.	DEVICE 3 button & H1~H4 indicator	Press the button to switch between Hosts(H1~H4) circularly for device 3. When the host is selected, the corresponding led will light up.
6.	DEVICE 4 button & H1~H4 indicator	Press the button to switch between Hosts(H1~H4) circularly for device 4. When the host is selected, the corresponding led will light up.

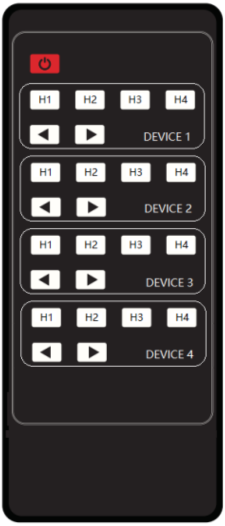
Operation Controls and Functions

Rear Panel



No.	Name	Function Description
1.	TCP/IP	100M LAN port for Web GUI and TCP/IP control, connected to a router or network switch.
2.	RS-232	Connect to a PC or control system for RS-232 control.
3.	IR EXT	Connect the included IR Receiver cable. If the IR signal receiving window of the unit is blocked or the unit is installed in a closed area out of infrared line of sight, the IR receiver cable can be inserted to the "IR EXT" port to receive the IR remote signal.
4.	HOST 1~4	Connect to a PC or Laptop through the USB port.
5.	DEVICE 1~4	USB-A device ports, with output power up to 5V/1.5A. Connect USB devices such as USB drives, keyboard or mouse. Note: The USB 3.2 Gen 1 ports do not support DisplayPort™ Video (DP Alt Mode) or Power Delivery (PD).
6.	DC 24V	Connect the included power adapter.

IR Remote & IR Cable Pin Assignment



⏻:
Power on the Matrix or set it to standby mode.

DEVICE 1:
Press H1\H2\H3\H4 button to select the host for DEVICE 1.

DEVICE 2:
Press H1\H2\H3\H4 button to select the host for DEVICE 2.

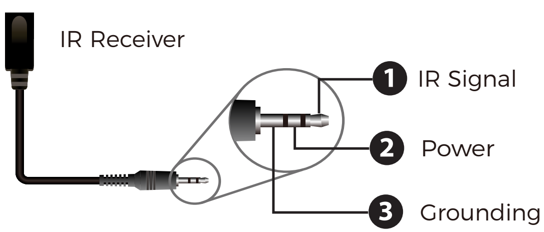
DEVICE 3:
Press H1\H2\H3\H4 button to select the host for DEVICE 3.

DEVICE 4:
Press H1\H2\H3\H4 button to select the host for DEVICE 4.

◀ ▶: Select the last or next signal input source.

IR Cable Pin Assignment

IR RECEIVER



Web GUI User Guide

The Matrix can be controlled by Web GUI. The operation method is shown below:

Step 1: Get the current IP Address.

The default IP address of the Matrix is 192.168.0.100. You can get the current IP address via an RS-232 command. Send the command *"get ip addr"* through an ASCII Command tool, then you'll get the feedback information as shown below:

```
IP Mode: Static
IP: 192.168.0.100
Subnet Mask: 255.255.255.0
Gateway: 192.168.0.1
TCP/IP port=8000
Telnet port=23
Mac address: 00:1c:91:03:80:01
```

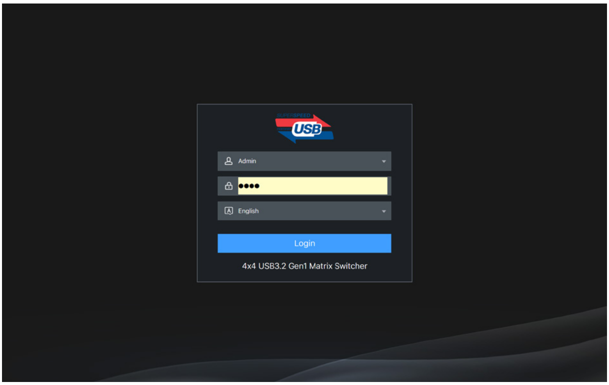
IP:192.168.0.100 in the above figure is the IP Address of the Matrix (the IP address is variable, depending on what the specific machine returns).

For the details of the RS-232 commands, please refer to Page 16.

Step 2: : Connect the TCP/IP port of the Matrix to a PC with a LAN cable, and set the IP address of the PC to be in the same network segment with the Matrix.

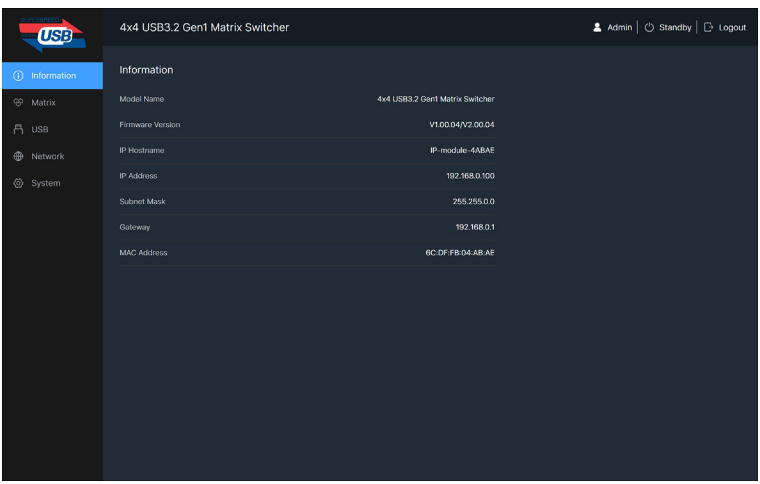
For example, if the IP address of the Matrix is 192.168.62.106, the IP address of the PC must be set to 192.168.62.104/105/107; but not exactly the same as the Matrix.

Step 3: Enter the IP address of the Matrix into your browser on the PC. After entering the Web GUI page, there will be a Login page, as shown below:



Select the username “Admin”, enter the default password “1234””, select the desired language, and then click “Login”. The web page provides the following tabs for navigating the interface:

■ Information Page

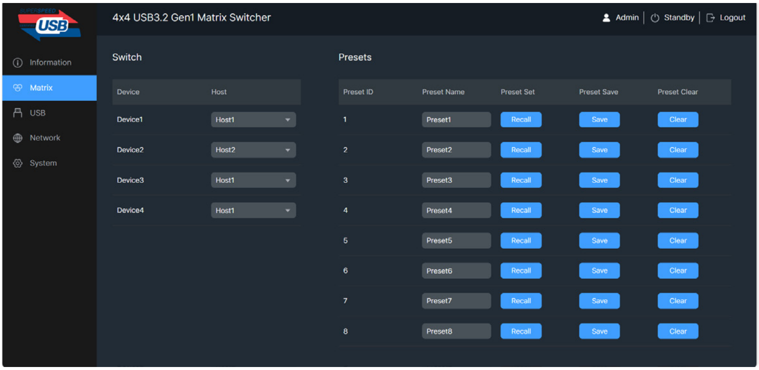


The Information page provides basic information about the Model, the installed firmware version and the network settings of the device.

The buttons at the top right of the web interface are always available:

- Clicking the Logout button will log out the current user.
- Clicking the Standby button will set the matrix to standby mode.

■ Matrix Page



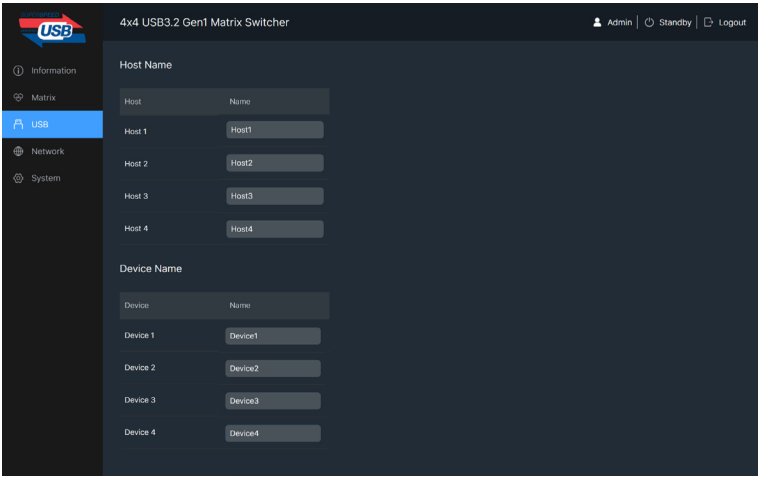
You can do the following operations on the Matrix page:

① **Switch:** Select the host for device1/2/3/4.

② **Presets:** Recall, save and clear the presets. 8 presets are allowed to be set.

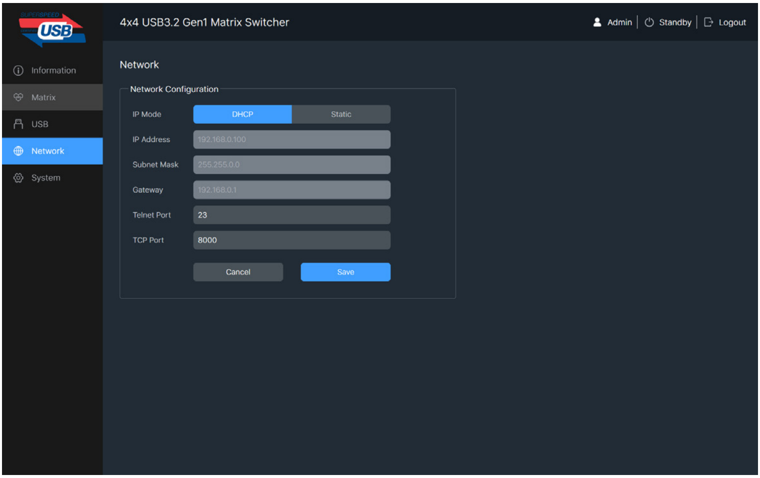
For example, a group of devices and hosts is configured, click the “Save” button to save this as a preset and name it “Preset1” or other. Then you can recall or clear it by clicking Recall or Clear if needed.

■ USB Page



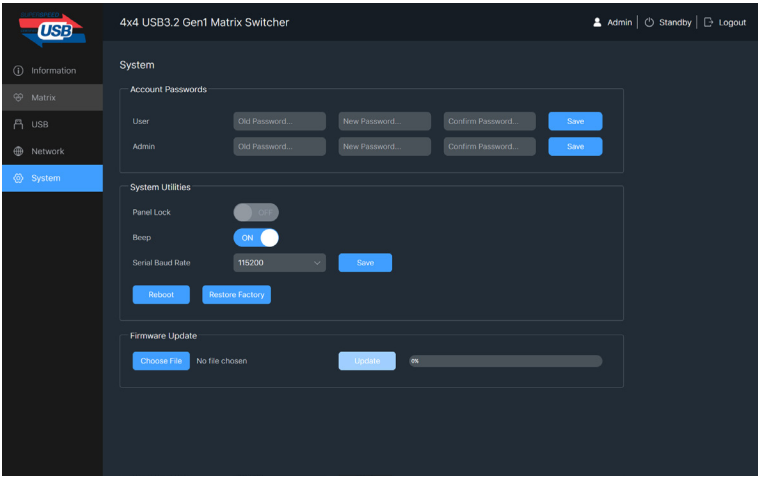
You can rename each host or device on the USB page.

■ Network Page



You can modify the IP Mode/IP Address/Gateway/Subnet Mask/Telnet Port/ Domain Name as required on the Network page. Click "Save" to save the settings, and then it will come into effect.
If the Mode is "DHCP", it will search and fill the IP Address assigned by the router automatically. You can't modify it now.
If the Mode is "Static", you can set manually the IP Address/Gateway/Subnet/ Telnet Port as required.

■ System Page



You can do the following operations on the System page:
① **Account Passwords:** Enter the correct Old Password, New Password, and Confirm Password, and then click "Save".
Note: Input rules for changing passwords:
(1) The password can't be empty.
(2) New Password can't be the same as Old Password.
(3) New Password and Confirm Password must be the same.

② **System Utilities:**

Panel Lock: Enable or disable the buttons on the front panel.

Beep: Turn on or turn off the beep.

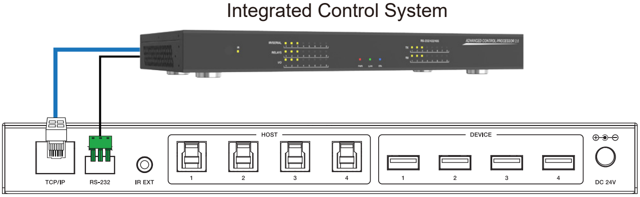
Serial Baud Rate: Set and save the baud rate: 4800/9600/19200/38400/115200/57600.

You can also reboot or restore to factory settings by clicking the corresponding buttons.

③ **Firmware Update:** Upload the firmware update file to update the device.

RS-232 Commands

The product also supports RS-232 command control. Connect the RS-232 port of the matrix to a control system or a PC. The connection method is as follows.



Then, open a Serial Command tool on the PC and send ASCII command to control the Matrix. The ASCII command list is below.

ASCII Command				
Serial port protocol. Baud rate: 115200 (default), Data bits: 8bit, Stop bits:1, Check bit: 0				
x,y,z, XXX are parameters				
Command Code	**Function Description**	**Example**	**Feedback**	**Default**
System Setting				
?	List all commands	?		
help	List all commands	help		
status	Get device current status	status	get the unit all status: power, beep, USB crosspoint, network status	
get model	get device model	get model	4x4 USB3.2 Gen1 Matrix	
get version	get firmware version	get version	mcu fw version :1.00.00 web gui version :1.00.00	
set power x	power on/off the device(z=0~1)	set power x	power on system initializing... mcu fw version :1.00.00 web gui version :1.00.00 initialization finished!	power 1

Command Code	Function Description	Example	Feedback	Default
System Setting				
get power	get current power state	get power	power on /power off	
set beep x	enable/disable buzzer function,z=0~1(z=0 beep off, z=1 beep on)	set beep 0	beep on beep off	beep off
get beep	get buzzer state	get beep	beep on / beep off	
set lock x	lock/unlock front panel button,z=0~1(z=0 lock off,z=1 lock on)	set lock 0	panel button lock on panel button lock off	panel button lock off
get lock	get panel button lock state	get lock	panel button lock on/off	
set baud rate x	set RS232 baudrate x=1~6 (1:115200, 2:57600, 3:38400, 4:19200, 5:9600, 6:4800)	set baud rate 1	baudrate:115200	baudrate: 115200
get baud rate	get the serial port baud rate of rs02 module	get baud rate	baudrate:115200	
reboot	reboot the device	reboot	reboot... system initializing... mcu fw version: 1.00.00 web gui version: 1.00.00 initialization finished!	
reset	reset to factory defaults	reset	reset to factory defaults system initializing... mcu fw version: 1.00.00 web gui version: 1.00.00 initialization finished!	
USB Matrix Setting				
set device x in host y	route usb device x to host y (x=1~4, y=1~4) x=1. device 1 x=2. device 2 x=3. device 3 x=4. device 4 y=1. host1 y=2. host2 y=3. host3 y=4. host4	set device 1 in host 1	set device 1 in host 1	device 1->host 1 device 2->host 1 device 3->host 1 device 4->host 1

RS-232 Commands

Command Code	Function Description	Example	Feedback	Default
USB Matrix Setting				
get device x in host	get device x selected host(x=1~4)	get device 1 in host	device 1 in host 3	
set save preset x	Save matrix state of all device and the host to preset z, z=1~8	set save preset 1	save to preset 2: device1 connect to host1 device2 connect to host1 device3 connect to host1 device4 connect to host1	no preset
set recall preset x	Call saved preset z scenarios, z=1~8	set recall preset 1	recall from preset 1: device1 connect to host1 device2 connect to host1 device3 connect to host1 device4 connect to host1	no preset
set clear preset x	Clear stored preset z scenarios, z=1~8	set clear preset 1	clear preset 1	no preset
get preset x	Get preset z infomation, z=1~8	get preset 1	preset 1: device 1->host 1 device 2->host 2 device 3->host 3 device 4->host 4	
Network Setting				
get ipconfig	get the current ip configauration	get ipconfig	ip mode: static ip: 192.168.0.100 subnet mask: 255.255.255.0 gateway: 192.168.0.1 tcp/ip port=8000 telnet port=23 mac address: 00:1c:91:03:80:01	
get mac addr	get network mac address	get mac addr	mac address: 00:1c:91:03:80:01	
set ip mode z	set network ip mode to static ip or dhcp, z=0~1 (z=0 static, z=1 dhcp)	set ip mode 0	set ip mode:static. (please use "s net reboot!" command or repower device to apply new config!)	ip mode:dhcp
get ip mode	get network ip mode	get ip mode	ip mode: static	

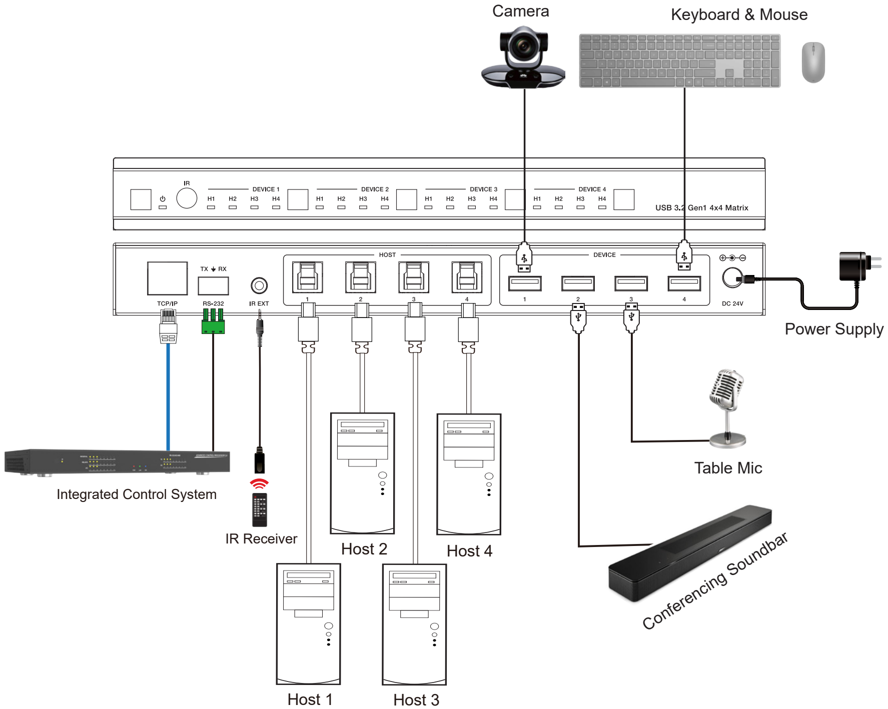
RS-232 Commands

Command Code	Function Description	Example	Feedback	Default
Network Setting				
set ip addr xxx.xxx.xxx.xxx	set network ip address	set ip addr 192.168.0.100	set ip dress: 192.168.0.100 (please use "s net reboot!" command or repower device to apply new config!) dhcp on, device can't config static address, set dhcp off first.	ip address: 192.168.0.100
get ip addr	get network ip address	get ip addr	ip address: 192.168.0.100	
set subnet xxx.xxx.xxx.xxx	set network subnet mask	set subnet xxx.xxx.xxx.xxx	set subnet mask: 255.255.255.0 (please use "s net reboot!" command or repower device to apply new config!) dhcp on, device can't config subnet mask, set dhcp off first.	subnet mask: 255.255.0.0
get subnet	get network subnet mask	get subnet	subnet mask: 255.255.255.0	
set gateway xxx.xxx.xxx.xxx	set network gateway	set gateway 192.168.0.1	set gateway: 192.168.0.1 (please use "s net reboot!" command or repower device to apply new config!) dhcp on, device can't config gateway, set dhcp off first.	gateway: 192.168.0.1
get gateway	get network gateway	get gateway	gateway:192.168.0.1	
set tcp/ip port x	set network tcp/ip port (x=1~65535)	set tcp/ip port 8000	set tcp/ip port:8000	tcp/ip port: 8000
get tcp/ip port	get network tcp/ip port	get tcp/ip port	tcp/ip port:8000	
set telnet port x	set network telnet port(x=1~65535)	set telnet port 23	set telnet port:23	telnet port:23

RS-232 Commands

Command Code	Function Description	Example	Feedback	Default
Network Setting				
get telnet port	get network telnet port	get telnet port	telnet port:23	
set net reboot	reboot network modules	set net reboot	network reboot... ip mode: static ip: 192.168.0.100 subnet mask: 255.255.255.0 gateway: 192.168.0.1 tcp/ip port=8000 telnet port=23 mac address: 00:1c:91:03:80:01	

Application Example



USB 3.2 Gen 1 4x4 Matrix

UKM-404

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